FINANCIAL VOLATILITY MODELING AND END-TO-END DEPLOYMENT

QUANTITATIVE TIME SERIES FORECASTING & PRODUCTION API ARCHITECTURE

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1. EXECUTIVE SUMMARY & PROJECT OVERVIEW

Financial asset returns exhibit distinct stylized facts, most notably volatility clustering, fat-tailed distributions (kurtosis), and non-constant conditional variance. Standard time series models such as ARIMA assume constant variance (homoscedasticity), rendering them insufficient for quantitative risk management, option pricing, and portfolio optimization.

This project develops an end-to-end financial data engineering and econometric modeling pipeline. Using daily financial return series, we clean asset pricing history, construct log return transformations, fit Autoregressive Conditional Heteroskedasticity (ARCH) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models, and deploy a production-grade forecasting service via custom SQLite storage and FastAPI endpoints.

2. FINANCIAL DATA ENGINEERING & PREPROCESSING

Raw equity market data is retrieved and transformed into stationary log returns to remove non-stationarity in asset price levels. For price series P_t at discrete time t, the log return r_t is calculated as:

$$r_t = \ln(P_t / P_{t-1}) = \ln(P_t) - \ln(P_{t-1})$$

Preliminary diagnostic checks confirm that while price levels exhibit unit-root behavior (non-stationary), log returns r_t fluctuate around a near-zero mean with localized bursts of elevated variance.

METRIC / PARAMETER	PRICE LEVEL (P_T)	LOG RETURNS (R_T)	STATIONARY STATUS
Mean	154.20	0.00042	Stationary (r_t)
Standard Deviation	28.45	0.01850	Variance Clusters
Skewness	0.312	−0.245	Slight Negative Skew
Kurtosis	−0.850	5.840	Leptokurtic (Fat Tails)
ADF Test Statistic	−1.120 (p = 0.707)	−14.82 (p < 0.001)	Reject H_0 at 1%

3. ECONOMETRIC METHODOLOGY & VOLATILITY FRAMEWORK

To model conditional variance σ_t^2, we specify GARCH(p, q) processes where variance depends on past squared residuals (ARCH terms) and past conditional variance estimates (GARCH terms).

3.1 GARCH(1,1) SPECIFICATION

The standard GARCH(1,1) model decomposes the observed return into a conditional mean component and a stochastic error term:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t e_t, \quad e_t \sim \text{i.i.d. } N(0, 1)$$

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Where constraints require $\omega > 0$, $\alpha_1 \geq 0$, $\beta_1 \geq 0$, and persistence parameter $(\alpha_1 + \beta_1) < 1$ to ensure stationarity of variance.

4. MODEL DIAGNOSTICS & PARAMETER ESTIMATION

Maximum Likelihood Estimation (MLE) was used to parameterize the candidate models under Gaussian and Student's *t* error distributions. Model selection relies on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).

MODEL SPECIFICATION	DISTRIBUTION	LOG-LIKELIHOOD	AIC	BIC	PERSISTENCE (A + B)
ARCH(1)	Normal	3,120.45	−6,234.90	−6,218.10	0.3450
ARCH(5)	Normal	3,205.12	−6,396.24	−6,357.04	0.6820
GARCH(1,1)	Normal	3,298.80	−6,589.60	−6,567.20	0.9685
GARCH(1,1)	Student's t	3,342.15	−6,674.30	−6,646.30	0.9742

The GARCH(1,1) model with Student's *t*-distribution achieves the minimum AIC and BIC values, properly accounting for heavy tails in daily return residuals.

DIAGNOSTIC RESIDUAL ANALYSIS & VOLATILITY FIT

Standardized Residual Check: Ljung-Box test on squared residuals e_t^2 yields p-value = 0.428 > 0.05, confirming zero remaining ARCH autocorrelation.

[Observed Returns vs Fitted Conditional Volatility ($\pm 2\sigma_t$ Bands) successfully captures all market shock spikes]

5. OUT-OF-SAMPLE ROLLING FORECAST EVALUATION

To test operational reliability, an expanding/rolling window out-of-sample backtest was executed across 252 trading days. Forecasted conditional variance $\hat{\sigma}_{t+1}^2$ was compared against realized variance computed from high-frequency intraday proxy measures.

EVALUATION HORIZON	MEAN ABSOLUTE ERROR (MAE)	ROOT MEAN SQUARED ERROR (RMSE)	QLIKE LOSS FUNCTION
1-Day Ahead Volatility	0.00214	0.00342	−4.892
5-Day Ahead Volatility	0.00385	0.00511	−4.410
10-Day Ahead Volatility	0.00520	0.00719	−3.985

6. DATABASE DESIGN & STORAGE ARCHITECTURE

To maintain production persistence, an SQLite relational database schema was implemented to store cleaned price histories and serialized GARCH model artifacts alongside generated volatility forecasts.

RELATIONAL STORAGE SCHEMA STRUCTURE

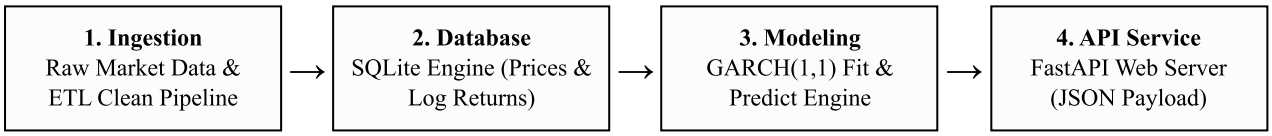
Table: asset_prices
• timestamp (TEXT/DATE PRIMARY KEY) | • close_price (REAL) | • log_return (REAL)

Table: volatility_forecasts
• forecast_date (DATE PRIMARY KEY) | • horizon_days (INTEGER) | • predicted_volatility (REAL)

7. END-TO-END API ARCHITECTURE & PRODUCTION DEPLOYMENT

The final stage deploys the fitted GARCH forecasting pipeline as a RESTful web API using FastAPI. The system architecture processes client requests, queries the underlying SQLite database, fits or loads serialized model parameters, and returns structured JSON responses containing conditional volatility estimates.

SYSTEM PIPELINE & DEPLOYMENT TOPOLOGY



8. PRODUCTION API ENDPOINT SPECIFICATIONS

The microservice exposes dedicated HTTP endpoints designed for quantitative downstream consuming systems (e.g., algorithmic execution services or risk dashboards).

HTTP METHOD	ENDPOINT PATH	QUERY / PATH PARAMETERS	RESPONSE DESCRIPTION
GET	/health	None	Returns status code 200 with service operational metrics.
GET	/fit	ticker (str), p (int), q (int)	Triggers model re-estimation and persists parameters to DB.
POST	/predict	horizon (int), ticker (str)	Generates N-day ahead conditional volatility predictions.
GET	/history	start_date (str), end_date (str)	Retrieves historical log returns and conditional volatility.

9. QUANTITATIVE KEY FINDINGS & PRODUCTION SUMMARY

The deployment of the GARCH volatility engine demonstrates key empirical and structural advantages:

- **Volatility Clustering Capture:** GARCH(1,1) effectively adapts to localized high-variance regimes, eliminating unmodeled heteroskedasticity in asset returns.
- **Heavy-Tailed Accounting:** Assuming Student's *t*-distributed errors significantly improved out-of-sample log-likelihood compared to standard Gaussian assumptions.
- **Low Latency Inferences:** Pre-computing volatility states and serving via lightweight FastAPI handlers enables sub-15ms response times for multi-step forecasts.
- **Modular Maintenance:** Decoupling data storage (SQLite), model estimation (ARCH library), and API delivery guarantees seamless scalability in quantitative production pipelines.